

SUMMARY

In my academic journey as a software engineering student at Karachi University (UBIT), I've dedicated myself to the world of data science. Over the past years, I have gained practical experience in various areas such as Machine Learning, Deep Learning, Computer Vision, and Generative AI. I aim to utilize my expertise, skills, and ability to work effectively in a team-based environment to drive innovation.

EXPERIENCE

Paysys Labs Pvt. Ltd. (Karachi)

Associate Data Scientist (Full-Time) • July. 2024 - Present

- Worked on computer vision based biometric projects utilizing object detection, segmentation, ocr, and enhanced fingerprint quality.
- RnD on liveness detection techniques.
- Fine-tuned large language models (LLMs) using company-specific datasets leveraging RAG.
- Containerized vision models with Docker and served them via REST APIs.

Folio3 Pvt. Ltd. (Karachi)

Artificial Intelligence Intern • Jul. 2023 - Sept. 2023

- Contributed to Folio3's machine learning projects, worked on vector databases, and embeddings, and created feedback loop pipelines for the model.

Proxima AI (ICCBS-KU, Karachi)

Data Science Intern • Oct. 2022 - Mar. 2023

- Made dashboards for better data access, gathered data from different places using web scraping, worked on machine learning models, and created custom DL models.

EDUCATION

Karachi University - UBIT

Bachelor of Science in Software Engineering (BSSE)
2020 – 2023

Government Inter Boys College

Intermediate
2018 – 2019

CERTIFICATIONS

Machine Learning Specialization (Coursera)

Deep Learning Specialization (Coursera)

NLP with Classification and Vector Spaces (Coursera)

Statistics For Business Analysis (365DataScience)

Optical Character Recognition MasterClass (Udemy)

Data Analysis With Python (FreeCodeCamp)

LangChain For LLM Application Development (DeepLearning.ai)

SKILLS

Domains

Machine Learning
Deep Learning
Computer Vision
NLP
Docker
LLMs
Aws Cloud
Azure Cloud
Vector Databases

Languages

Python
C
Bash
Html / Css
Sql

Libraries

TensorFlow
Pytorch
Ultralytics
Mediapipe
OpenCV
Keras
HuggingFace
NLTK
LangChain
LangGraph
Scikit Learn
Selenium
Flask
FastAPI
Librosa
Unsloth
Ollama
Streamlit
Gradio

PROJECTS

GenView (AI-Based Interview System)



Use Case: To assist human resource managers in filtering out high-performing candidates.

- LLM-driven interview system that collects interviewee data, poses questions, and assesses responses for accuracy.
- The user logs into the system and provides information such as experience, libraries they worked with, their expertise in each library's past projects, etc.
- The system uses Openai services, to process users' information and generate queries.
- Used vector database (Qdrant) to store answers by converting them into embeddings for similarity purposes.

Real-Time License Plate Extractor



Use Case: To track, identify, and log license plates along with the entry times of vehicles at a specific point.

- Developed a real-time system to capture and extract license plate information using computer vision.
- Trained custom YOLOv8 on license plate data.
- Utilized Keras OCR to extract text from the license plate.
- Optimized the pipeline for real-time performance, ensuring minimal latency in detection.

Image Style Transfer Paper Pytorch Implementation



Use Case: To blend an image and style provided by the user, producing a new image that mixes the two elements.

- Implemented neural style transfer using PyTorch based on the original research paper by Gatys et al.
- Utilized Convolutional Neural Networks (CNNs) to extract content and style features from images.
- Applied feature reconstruction from pre-trained VGG-19 to blend style and content effectively.
- Enhanced performance by tweaking hyperparameters.

Chat KU



Use Case: To help students at Karachi University find specific information about their department without having to go through a document.

- A fine-tuned chatbot was developed to answer questions regarding the university prospectus.
- An open-source tool was utilized to convert PDF data into a fine-tunable Alpaca format.
- Mistral-Nemo-Base-2407 was used as a base model for fine-tuning.
- Served fine-tuned model via gradio interface.

C41 - Ferrari (F1)



Use Case: To identify similarities between two or more cars and deploy models with high accuracy in the inference pipeline.

- Configured vector database, and utilized custom-trained CNN to create image embeddings of F1 cars for similarity purposes.
- Custom-trained YOLOv5 on the F1 cars dataset.
- Handled model conversion for TensorRT to optimize the performance of the model.
- Utilized NorFair library for the tracking of multiple objects using YOLOv5.
- Customized callback functions using Tensorflow.
- Created a feedback loop pipeline for experiment tracking and logging using custom callbacks.

Employee Handbook LLM



Use Case: To help employees at Paysys Labs find specific information related to employment, such as benefits, leaves, etc.

- A fine-tuned chatbot was developed to answer questions related to the Paysys Labs Employee Handbook.
- LLaMA 3.1 8B was used as a base model for fine-tuning.
- The Quantization Aware Training (QAT) approach was utilized to handle this project.
- A retrieval-augmented generation (RAG) architecture was implemented alongside the fine-tuned model to minimize hallucinations.

Biometric Verification System



Use Case: To integrate a biometric authentication system within PayApp, enabling users to confirm their identity securely.

- Custom-trained YOLOv5 models were utilized for detection and segmentation purposes on different datasets for the system.
- Developed a module for finger liveness to prevent users from gaming the system.
- Trained a classifier to detect spoofing attempts.
- R&D were conducted on finger liveness to enhance system security.

CSR System



Use Case: To assist merchants in resolving their queries and automatically log these issues as Jira tickets.

- Mistral-Nemo-Base-2407's function tool-calling ability was utilized to define functions for handling different product domains.
- Different product documents were vectorized and stored in a vector database.
- A retrieval-augmented generation (RAG) architecture was employed for the LLM agent to retrieve contextual information from the vector database.

Stable Diffusion (LoRA On Multiple Concepts)



Use Case: To generate images of multiple humans with character interactions using a single LoRA model.

- Captioned and assigned trigger words on images using an open-source model.
- Tried different methods such as dreambooth and open-source notebooks for training LoRA.
- Multiple Stable Diffusion extensions, such as ADetailer, ControlNet, and Latent Couple, were utilized to enable character interactions.